How-To Library

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How to configure assume roles for Amazon resources

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Abstract

This article describes how you can configure an assume role to access the Amazon Web Services (AWS) resources from the same or different AWS accounts.

Supported Versions

- Informatica® Cloud Application Integration Amazon S3 Connector
- Informatica® Cloud Application Integration Amazon SQS Connector

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Overview

You can use an assume role to allow the existing Identity and Access Management (IAM) users to access the Amazon Web Services (AWS) resources.

You can use an IAM role to access and manage the AWS resources and services. You can use IAM roles to delegate access to IAM users managed within your account, IAM users under a different AWS account, and an AWS service such as EC2.

Use case 1: Assume role with an IAM user in the same account

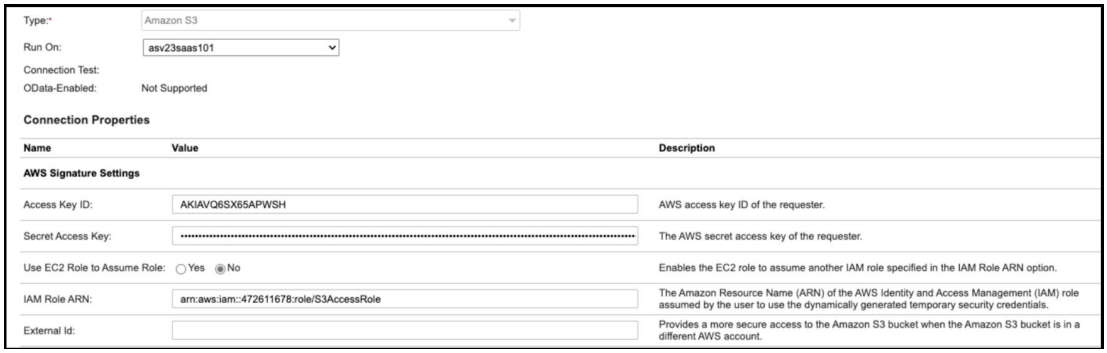
You can configure an assume role for an IAM user when the IAM user and the IAM role are in the same account. The IAM user has permission only to assume the role.

You must configure Cloud Application Integration and AWS to use the assume role.

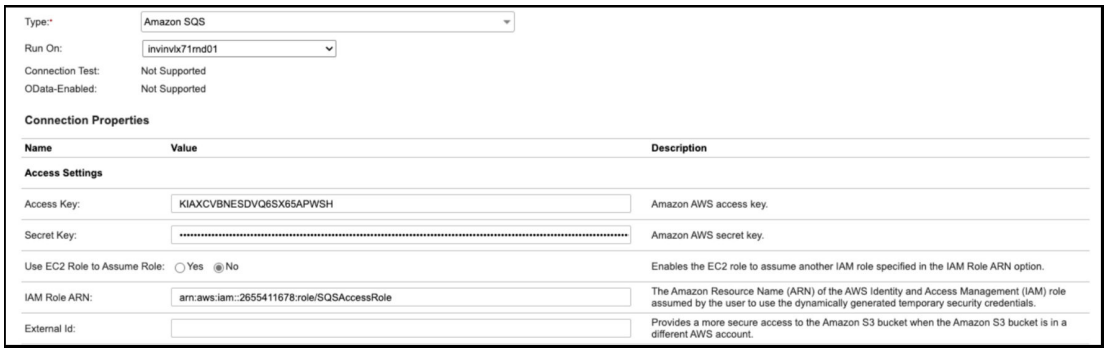
Configure the connection properties in Cloud Application Integration

Specify the access key, secret key, and the IAM Role ARN in the connection properties in Cloud Application Integration.

The following image shows the connection properties that you configure in Amazon S3 Connector for an account setup:



The following image shows the connection properties that you configure in Amazon SQS Connector for an account setup:

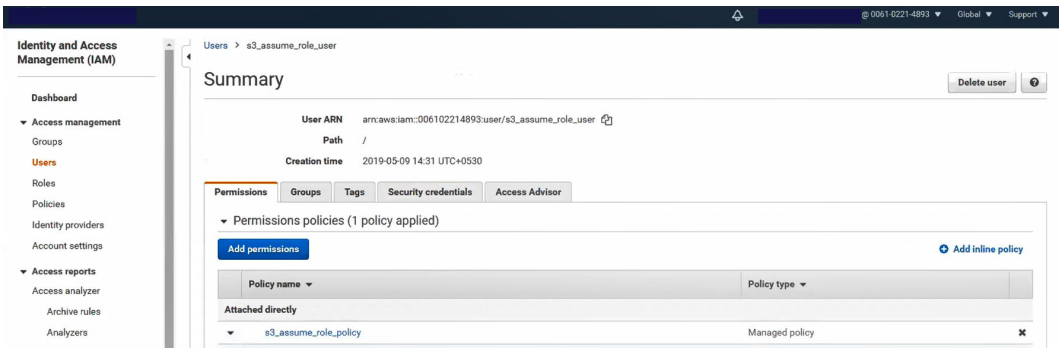


Configure assume role on the AWS console

Perform the following steps on the AWS console to configure assume role when the IAM user and the IAM role are in the same account:

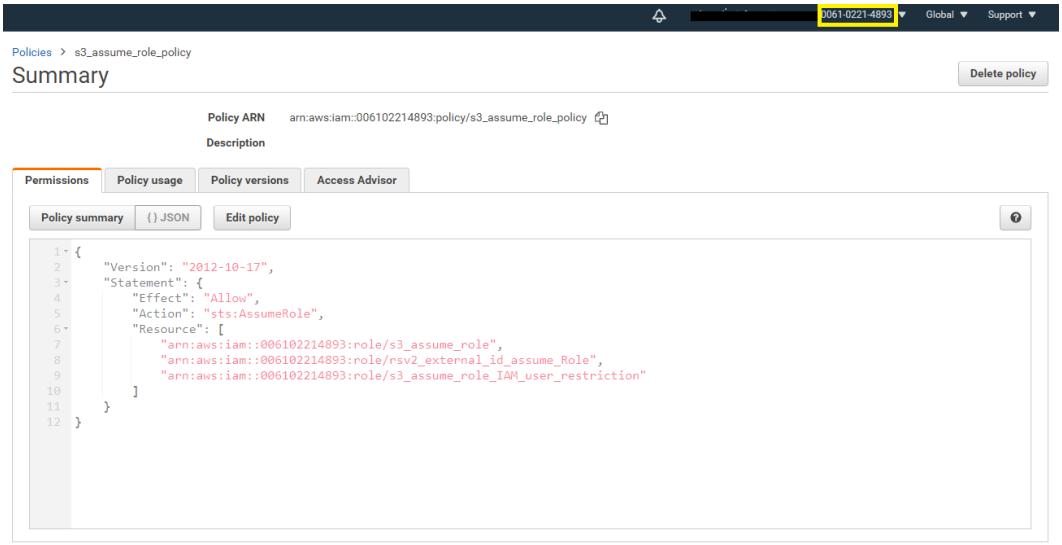
1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel. The **AWS Service** dashboard page appears.
3. Click **IAM**. The **Welcome to Identity and Access Management** page appears.
4. Click **Users** from the left panel.
   Create an IAM user and attach a policy to the IAM user.

   The following image shows the sample of how to configure an assume role on the AWS console:

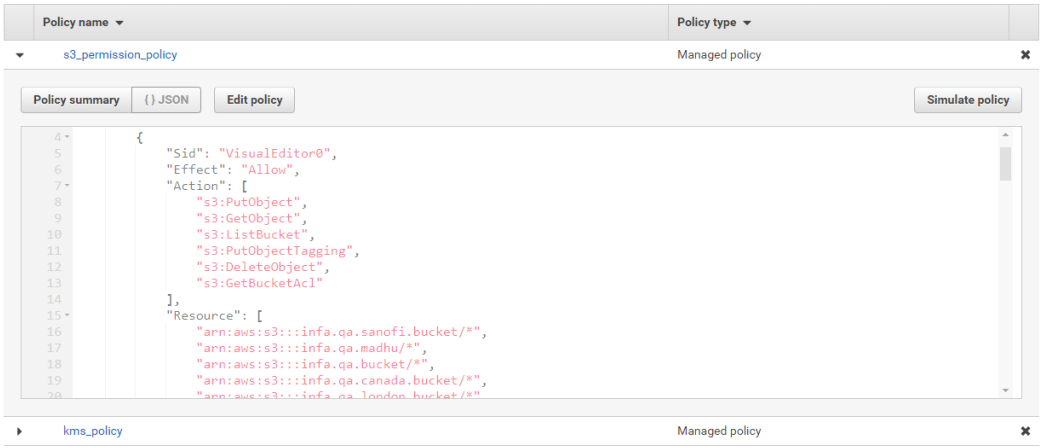


5. Click **Policies** from the left panel.
   The **Policies** page appears.

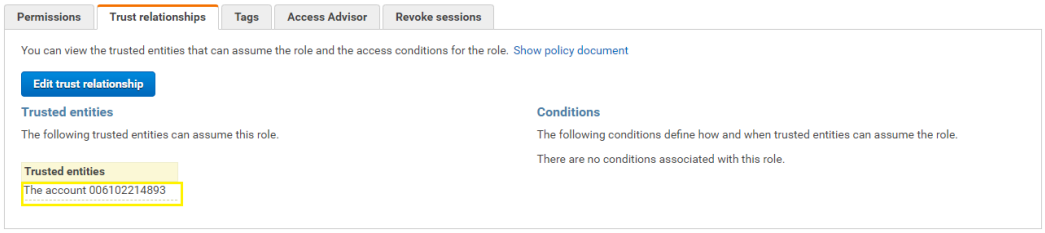
The following image shows a sample policy attached to the IAM user in account `0061-0221-4893` (account A):



6. Click **Roles** from the left panel.
The following image shows the policies attached to the IAM role:



7. Click the **Trust relationships** tab to define the trust relationship within the AWS account.
The following image shows that the user in account A is trusted to assume the role that you defined:



Use case 2: Assume role with an IAM user and IAM role in different accounts

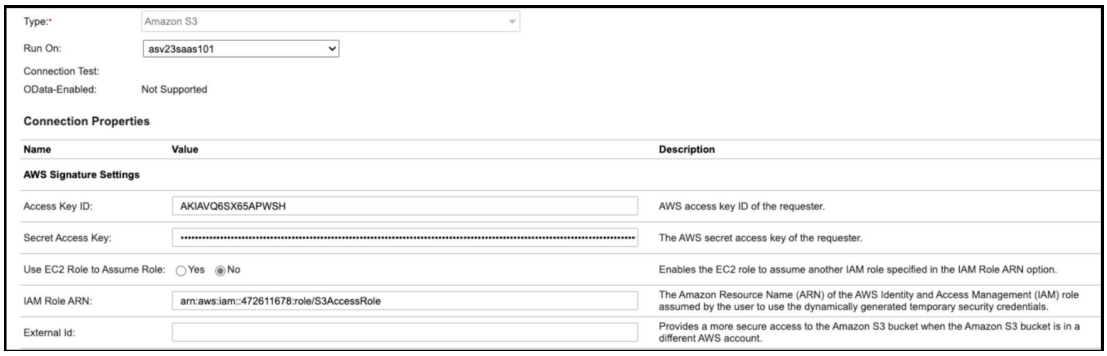
You can configure an assume role for an IAM user when the IAM user and the IAM role are in different accounts.

You must configure Cloud Application Integration and AWS to use the assume role.

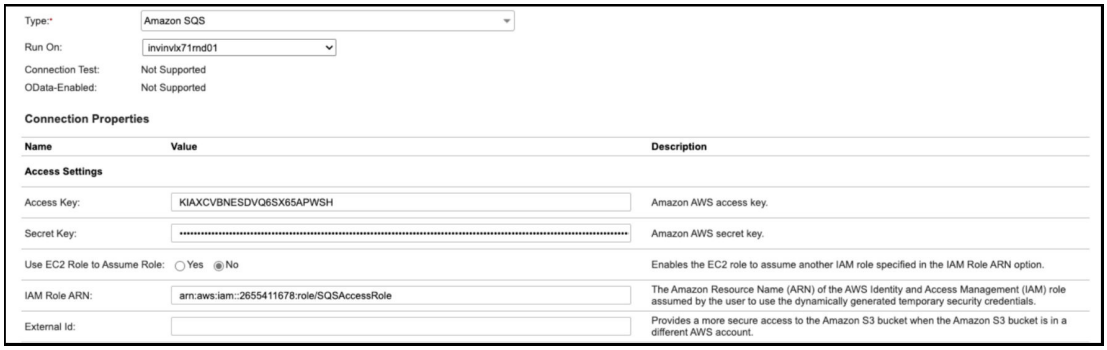
Configure the connection properties in Cloud Application Integration

Specify the access key, secret key, and the IAM Role ARN in the connection properties in Cloud Application Integration.

The following image shows the connection properties that you configure in Amazon S3 Connector for an account setup:



The following image shows the connection properties that you configure in Amazon SQS Connector for an account setup:



Configure assume role on the AWS console

You can configure an user in an account to assume a role in another account and access specific resources that belong to the account.

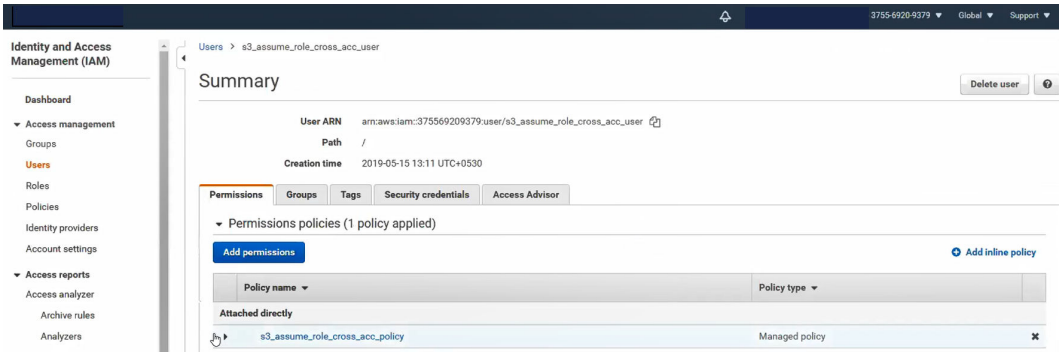
A user in account `3755-6920-9379` (account A) can assume a role in account `0061-0221-4893` (account B) to access specific resources of account B.

Perform the following steps on the AWS console to configure an assume role when the IAM user and the IAM role are in different accounts:

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel. The **AWS Service** dashboard page appears.
3. Click **IAM**. The **Welcome to Identity and Access Management** page appears.

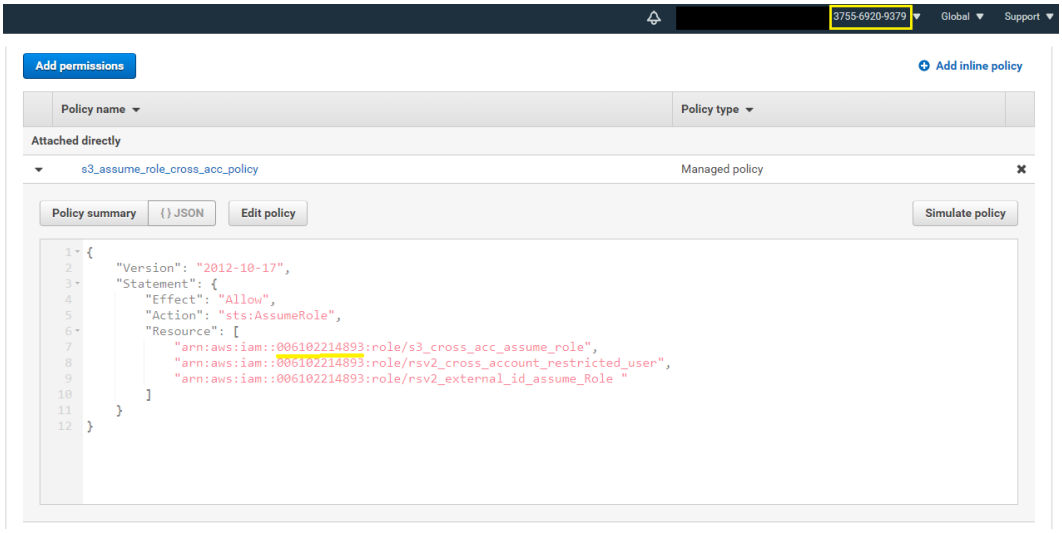
4. Click **Users** from the left panel.
   Create an IAM user and attach a policy to the IAM user.

   The following image shows the sample of configuring an assume role on the AWS console:



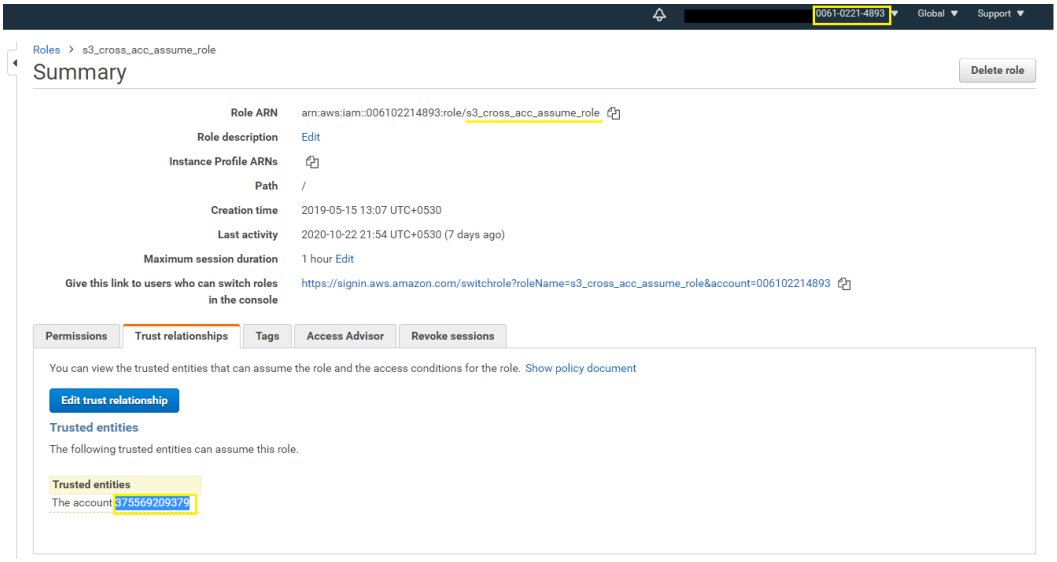
5. Click **Policies** from the left panel.
   The **Policies** page appears.

   The following image shows a sample policy attached to the IAM user in account A:



6. Define an IAM role in account B. Click **Roles** from the left panel. Configure the policies for the IAM role that you configured when the IAM user and the IAM role were in the same account.

7. Click **Trust relationships** tab to define the trust relationship within the AWS account.
   The following image shows that a user from account A is trusted to assume the role that you defined in account B:



Use case 3: External ID for a secure cross-account access

You can specify an external ID for a secure cross-account access to the Amazon S3 bucket when the Amazon S3 bucket is in a different AWS account. Similarly, you can specify an external ID for a secure cross-account access to the Amazon SQS bucket when the Amazon SQS bucket is in a different AWS account.

You must configure Cloud Application Integration and AWS to use the assume role.

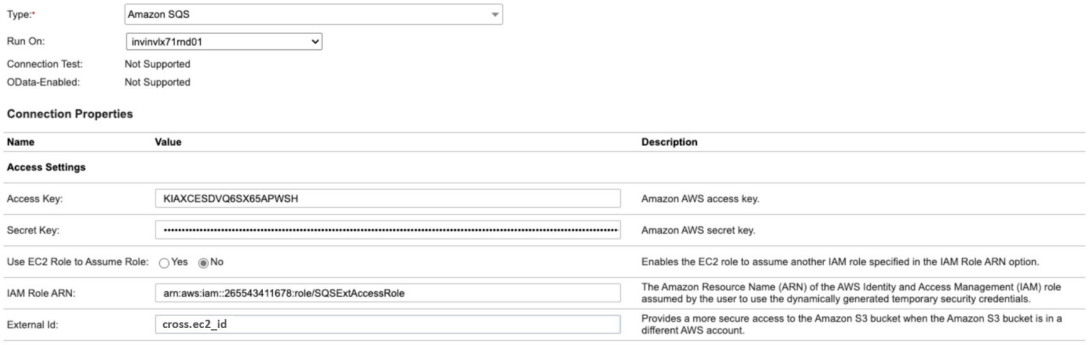
Configure the connection properties in Cloud Application Integration

Specify the IAM role ARN and the external ID in the connection properties in Cloud Application Integration.

The following image shows the connection properties that you configure in Amazon S3 Connector:



The following image shows the connection properties that you configure in Amazon SQS Connector:

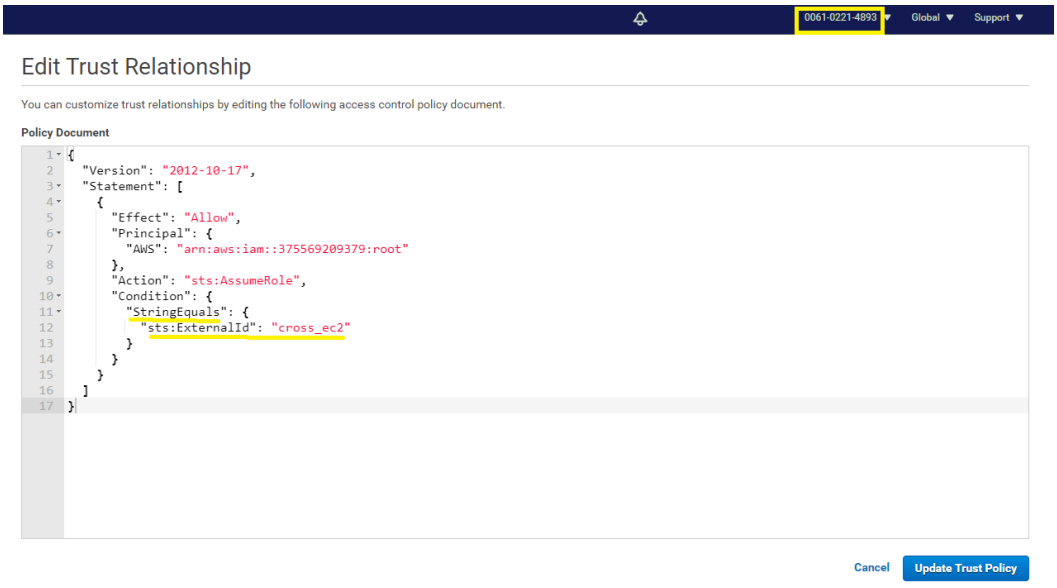


Configure assume role on the AWS console

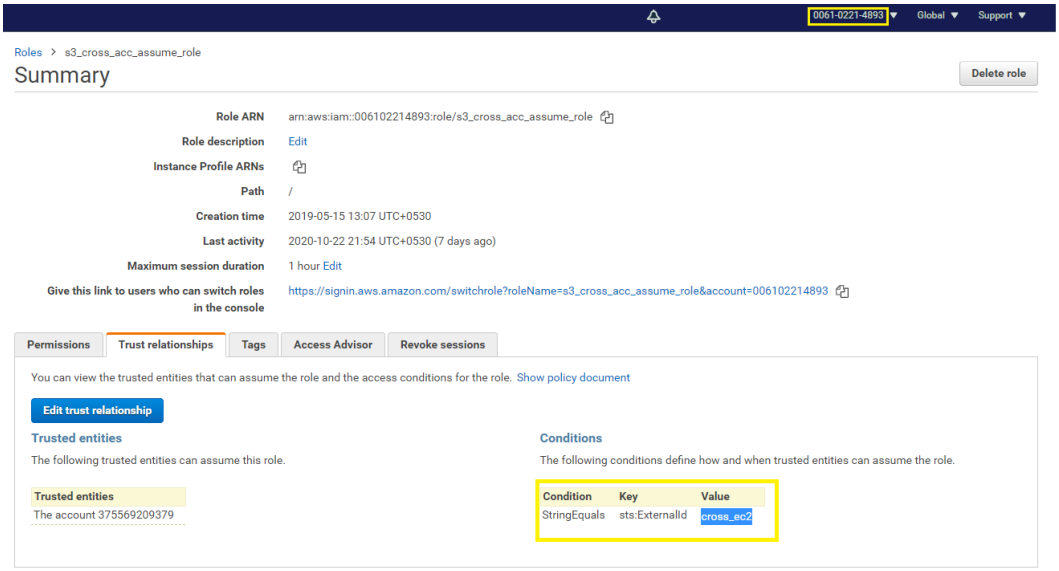
When an IAM user from account A tries to assume a role in account B, the IAM user needs to specify an external ID to be authenticated to assume this role even though you have defined the rules and policies for the IAM user and IAM role.

Perform the following steps on the AWS console to configure the assume role:

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel. The **AWS Service** dashboard page appears.
3. Click **IAM**. The **Welcome to Identity and Access Management** page appears.
4. Click **Policies** from the left panel. Configure the policies for the IAM user that you configure when the IAM user and the IAM role are in different accounts.
5. Click **Roles** from the left panel. Configure the policies for the IAM role that you configure when the IAM user and the IAM role are in different accounts.
6. Click the **Trust relationships** tab to view the trust relationship for the AWS account.
7. Click **Edit trust relationship** to define the trust relationship. The **Edit Trust Relationship** window opens.
8. Edit the policy and specify the conditions for the external ID.
   The following image shows the condition that you defined for the external ID:



9. Click **Update Trust Policy**.
The **Trust relationships** tab shows the external ID condition and the value that you specified.



Use case 4: Assume role using EC2

You can use an assume role for an Amazon EC2 role to access the AWS resources from the same or different AWS accounts. When you use an assume role with EC2, the Secure Agent must be on an EC2 container.

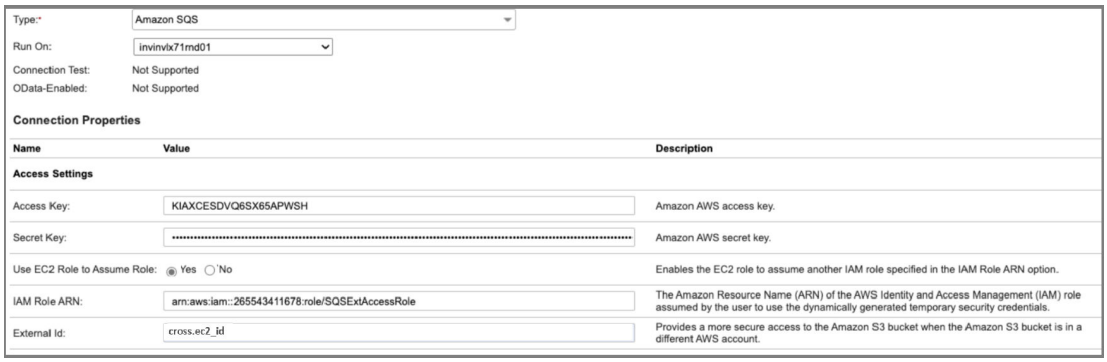
The Amazon EC2 role can assume another IAM role from the same or different AWS account without requiring a permanent access key and secret key.

You must configure Cloud Application Integration and AWS to use the assume role.

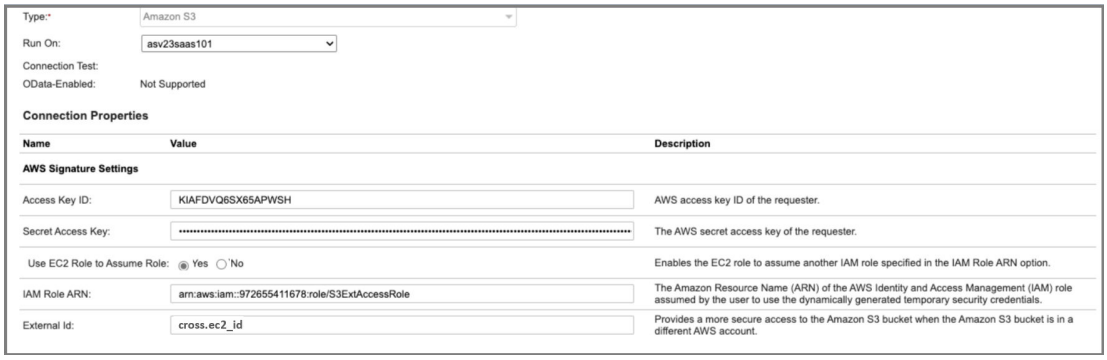
Configure the connection properties in Cloud Application Integration

Specify **IAM Role ARN** and set the **Use EC2 Role to Assume Role** property to **Yes** in the connection properties in Cloud Application Integration.

The following image shows the connection properties that you configure in Amazon SQS Connector:



The following image shows the connection properties that you configure in Amazon S3 V2 Connector:

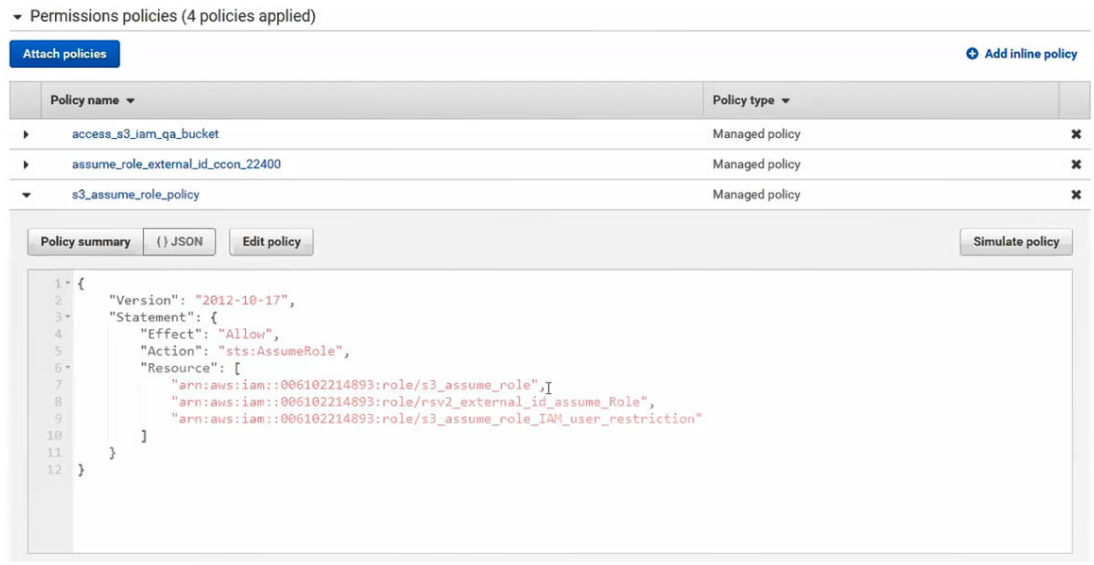


Configure assume role on the AWS console

The EC2 role assumes a role that an IAM user assumes when the IAM user and the IAM role are in the same account.

Create an EC2 instance and assign a policy to an EC2 role, which in turn has an assume role. Then, install the Secure Agent.

The following image shows the sample policies attached to an EC2 role:



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